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URTeC: 3723784

The Diminishing Returns of Extending Lateral Length Across Different Basins

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This paper was prepared for presentation at the Unconventional Resources Technology Conference held in Houston, Texas, USA, 20-22 June 2022.

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Abstract

Over the past decade, operators have significantly increased unconventional well performance through drilling longer laterals. While longer lateral length is known to benefit production, the exact efficiency of the additional length is a subject of much debate, in part due to the difficulties in separating the impact of longer laterals from upsized completions, interwell spacing, and geology. In this study, we use machine learning to disentangle the complex, nonlinear relationships to generate estimates of lateral length production efficiency across the major unconventional plays in the US.

We created a dataset for horizontal wells in six different basins, the Delaware, Denver-Julesburg, Eagle Ford-Gas Window, Eagle Ford-Oil Window, Midland, and Williston basins. For all regions, we built a basin-specific ensemble of decision tree models that utilized lateral length, completions, geology, and spacing as input features to predict three-year cumulative oil/condensate production. These models are then used to forecast a multitude of hypothetical wells placed throughout each basin.

For each basin, a lateral length efficiency curve is created using the relative performance degradation for increasing well lengths when comparing against the results from wells with 5,000 ft. This study reveals that increasing well lengths does not result in proportionate production gains. In fact, in some basins a 2-mile-long lateral only produces 85% of the normalized production of 1-mile-long laterals. We also find that the size of the completions job can influence the shape of the lateral length efficiency curve.

Oftentimes operators are planning for future wells with very long laterals without adjusting expectations on production per foot. However, for every basin in this study, we demonstrate there are varying magnitudes of diminishing return on production per lateral foot. Asset valuations need to take each PUD's length into account and adjust projected reserves accordingly. Similarly, for certain basins, planners will need to carefully weigh the costs and risks of drilling long laterals vs the outcome of drilling two separate wells.

Introduction

Since 2012, operators have increased average lateral length (LL) dramatically across all of the major unconventional plays, in order to increase well production and improve capital efficiency (Figure 1). However, predicting the production performance of longer wells oftentimes can either be quick or accurate.



Figure 1. Box plot of well lateral lengths per first production year grouped by basin.

Traditionally, understanding the performance of longer laterals has proven to be difficult. The simplest and quickest method is to find representative wells in an area, divide the identified wells' production by lateral length, and aggregate the transformed production history to generate a "per lateral length" scalar. An example of this quick analysis for all Delaware wells' 3 year cum oil per length vs lateral length is shown in Figure 2. Although a straight line fit through this scatter plot can provide high level trends, the uncertainty in performance is extremely high. For example, while the wells in Pecos County do exhibit the overall trend of longer laterals producing less oil per length, there is a trimodal distribution of lateral lengths (5,000, 7,500, and 10,000) with each bin having a high range of outcomes; a Wolfcamp A well with 5,000 ft could produce 20 bbl/ft or 60 bbl/ft (a ~200% difference).

This first approach is a hasty means to understand the diminishing returns of increasing lateral length, but there are many other well parameters that can influence this analysis. Do the wells have similar interwell spacing? Are the completions sizes relatively close to each other? Without accounting for more than just the geographic location of representative wells, the results will not show a concise trend. Han et al. (2021) created several well groups to analyze Delaware wells. While these modifications help with getting more concise trends, manually creating groups is highly dependent upon the range of parameters used to define each bin which introduces human bias.

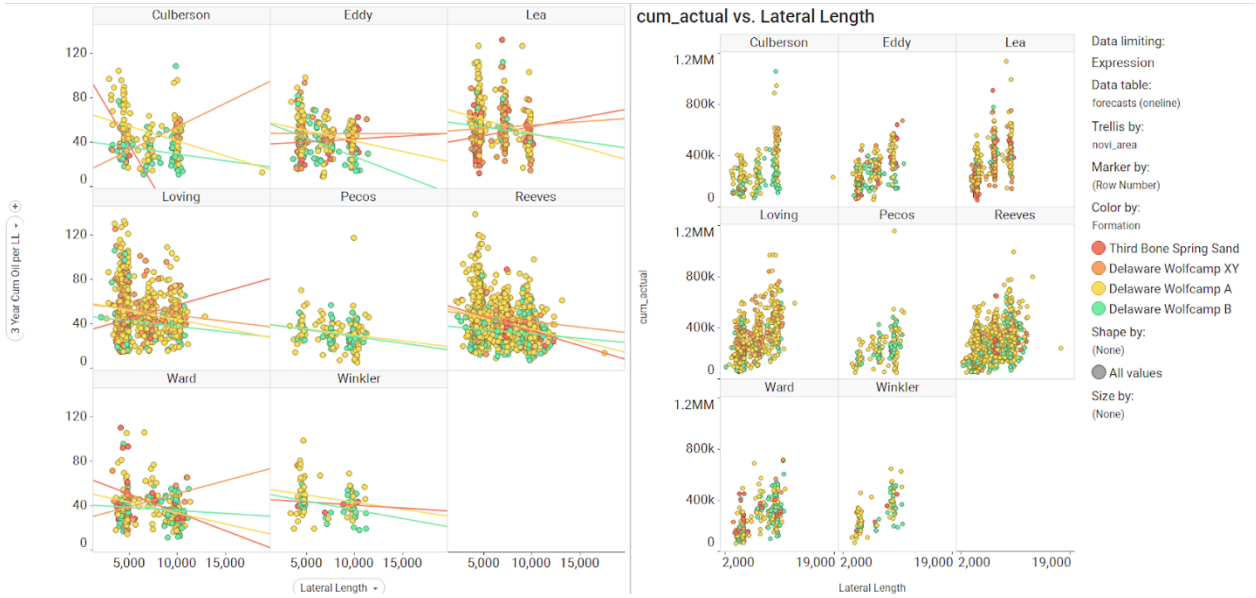


Figure 2. Actual 3 year per foot cum oil production in the Delaware basin. Overall trend shows diminishing returns for per foot production for longer wells. Additional well parameters create a wide range of outcomes and need to be taken into account.

Reservoir simulation approaches can provide a more detailed perspective. Pradhan & Xiong (2017, 2018) use a multitude of parameters and adjust well lengths to quantify the impacts of extending lateral length. However, creating a history-matched model for a statistically significant number of wells can be extremely time consuming.

Another approach to evaluating lateral length efficiency is to drill wells as part of a pilot program. By dedicating pads for testing different well lengths, operators can gain some understanding of efficiencies within a given geologic area. However, these development plans can include tens of wells, and therefore, have a significant capital exposure. Additionally, traditional trial and error techniques could take a few years before being able to provide insights while also forcing sub-optimal development of acreage used in test pads.

Machine Learning (ML) models, by contrast, excel at disentangling complex, nonlinear relationships, and provide a statistical perspective on the relationship between input and output variables. In this paper, we present a workflow that utilizes ML models to quickly isolate and test the impacts of lateral length on oil production across the main US unconventional plays.

Methods

We built machine learning models and conducted our analysis on horizontal wells in the Bakken, Denver-Julesburg, Eagle Ford-Gas Window, Eagle Ford-Oil Window, Midland, and Delaware basins. This section only outlines the steps used to conduct the Delaware basin analysis, however, similar workflows were followed for others.

We assembled a dataset of publicly available production and completions data. Using directional surveys, we calculated interwell spacing measurements. We also joined standard subsurface properties, such as water saturation, to each well location. We limited the dataset to wells with greater than 3,000 ft LL. For each basin, we trained a decision-tree based ML model to predict production with the aforementioned well parameters as predictive features. Final validation steps using SHAP values were conducted to

ensure that basin-wide production trends aligned with industry expectations (Lundberg et al., 2018; Cross et al., 2020).

We used the ML models to forecast the outcomes for hypothetical wells placed throughout each basin. This method allows us to design a series of development scenarios to control for independent variables. More specifically, we create dozens of pads and change lateral length across scenarios, and compare the forecast results at each location.

Each forecasted pad consisted of 3 wells in a single formation with interwell spacing set to 1,000 ft. All existing PDP wells were ignored from the prediction scenarios to account for undesired influences from parent spacing and depletion. Within each scenario, all hypothetical wells have the same lateral length value, however, across scenarios well lengths start from 5,000 and increase in 2,500 ft increments. 4 different fluid intensities that span from 30 to 60 bbls/ft are used. Proppant per length was also increased proportionally to maintain a constant proppant to fluid ratio of 0.8 lbs/gal.

In this analysis, we compare the 3 year per foot cum oil production for the interior well at each location. Figure 3 shows the comparison of the oil forecasts for 5,000 ft vs 10,000 ft scenarios for the Delaware Basin. The spatial plot of results indicate different areas within a basin will have different lateral length efficiencies.



Figure 3. Map of lateral length efficiencies for increasing well lengths from 5,000 ft to 10,000 ft across the Delaware basin. Areas colored red have a minimal change in per foot oil forecast when increasing the wells' lateral length. While blue points represent higher diminishing returns on production when extending wells.

Furthermore, to obtain basin-wide lateral length efficiency values, we record the median performance across all locations for each of the scenarios. Then, we analyze the scenarios with the same completion size and create a lateral length efficiency curve by dividing each lateral length scenario's median forecast by the median prediction of 5,000 ft wells. In other words, for each completion size, the scenarios with ~5,000 ft LL are assigned with the base efficiency of 100%; all efficiency metrics of longer lengths relate back to the production performance at 5,000 ft. Figure 4 shows the forecasted 3 year per foot cum oil production for all 16 scenarios and the resulting lateral length efficiency curve.

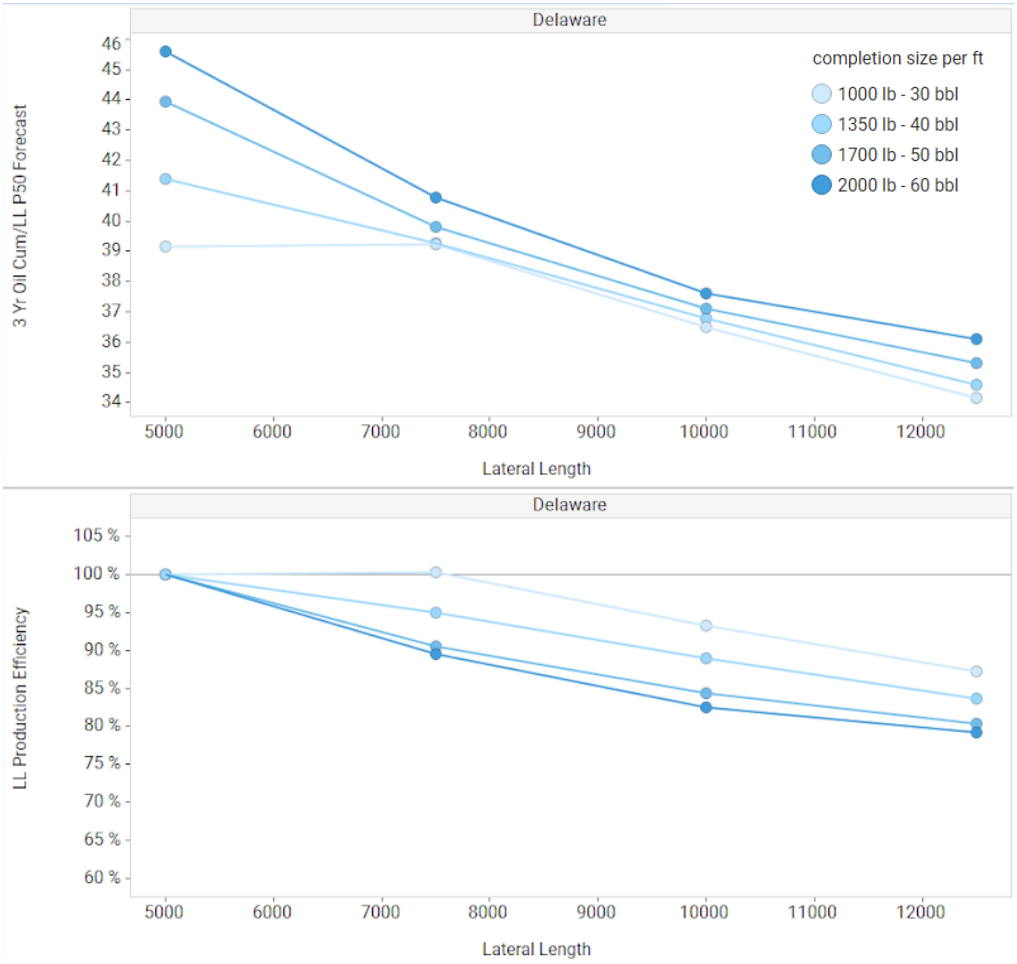


Figure 4. (Top) 3 year per foot cum oil production for all forecast scenarios. (Bottom) Transformed forecast lateral length efficiency curve with 5,000 ft wells set to 100%.

Although these basin-wide values are useful for generalization and easier for discussion purposes, it should be noted that lateral length efficiency is dependent on geographical location and geological conditions.

Results

Eagle Ford

The Eagle Ford basin was divided into two separate models, the Oil Window and the Gas Window. In both subsets, we observe that using less proppant and fluid will produce a lower amount of oil/condensate for Lower Eagle Ford wells, however, the lateral length efficiencies are better for smaller completion jobs. Additionally, in areas with higher GOR, we find higher lateral length efficiencies.

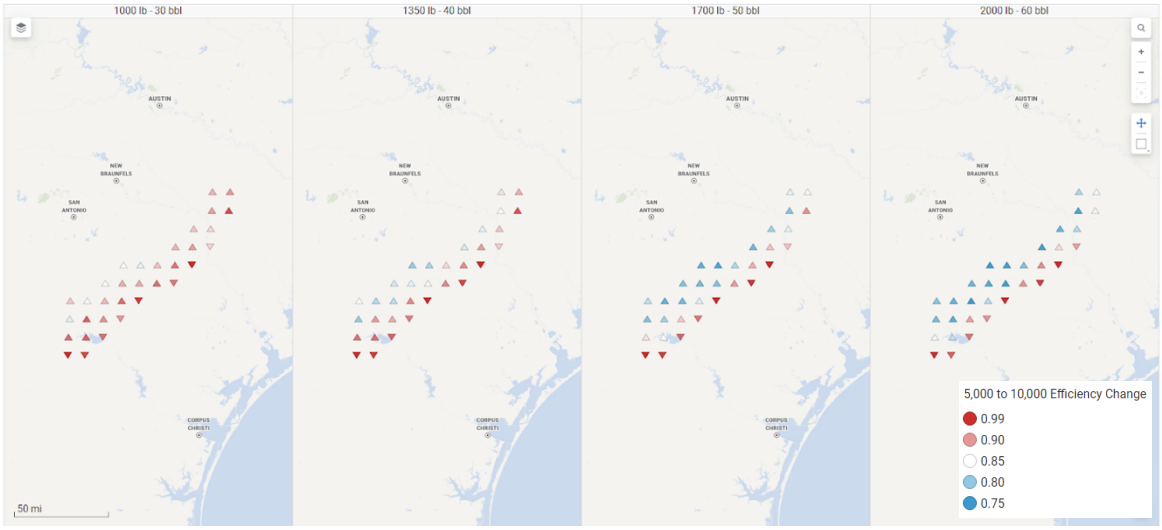


Figure 5. Lateral length efficiency for increasing from 5,000 ft to 10,000 ft in the Eagle Ford Oil and Gas Windows.

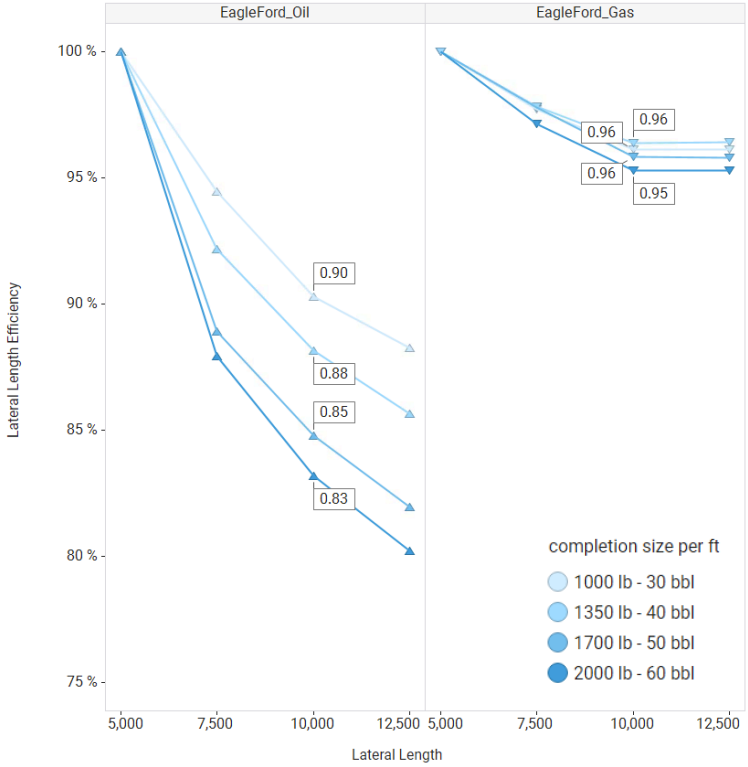


Figure 6. Lateral length efficiency curve for each completion design within the Eagle Ford.

Permian

In the Delaware Wolfcamp A, across all completion sizes, as lateral length increases, per foot oil production decreases. In the 2,000 lb/ft and 60 bbl/ft forecast set, wells with 10,000 ft LL have a 17% degradation in per foot production compared to the performance of wells with 5,000 ft LL and the same completion size. Interestingly, while using a smaller completion size reduces the total barrel of oil forecast, the lateral length efficiency curve has a shallower slope. In contrast, the Midland basin scenarios show much higher lateral length efficiencies for all completion sizes. The efficiency for these Wolfcamp A wells is at or above 95%.

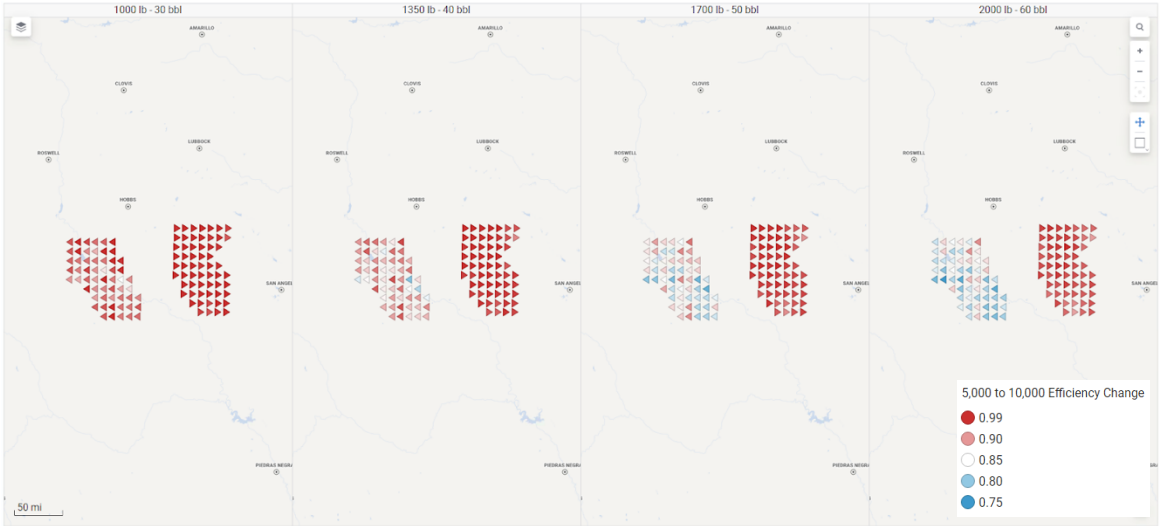


Figure 7. Lateral length efficiency for increasing from 5,000 ft to 10,000 ft in the Delaware and Midland.

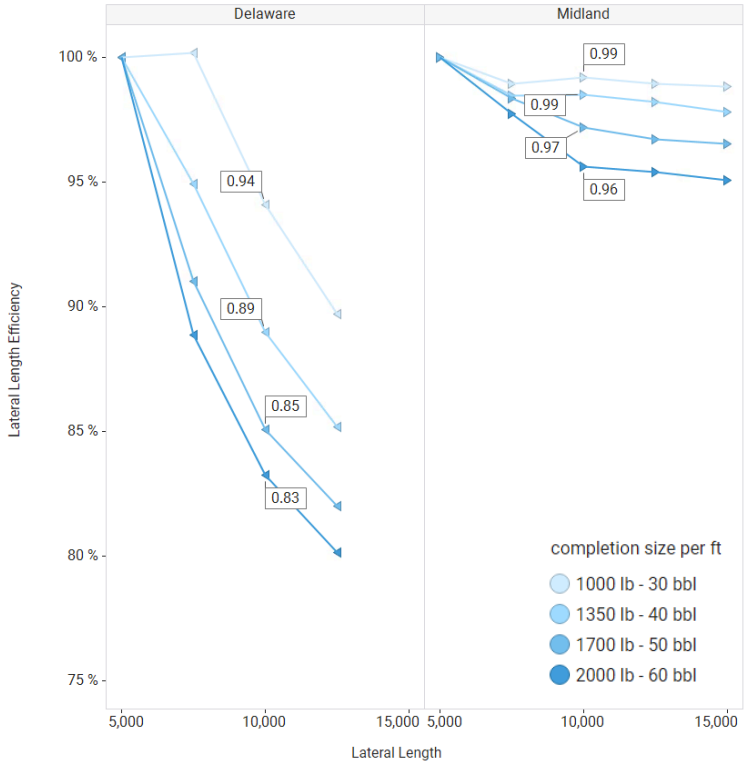


Figure 8. Lateral length efficiency curve for each completion design within the Delaware and Midland.

DJ & Williston

Wells in the DJ basin were the only ones to show that using more proppant and fluid can help lateral length efficiencies by a significant amount. The forecasted Niobrara B wells had up to a 11% change in efficiency when increasing completions size. While the forecasts for Middle Bakken wells also showed higher completions intensity would yield better lateral length efficiencies, the magnitude of impact is only 1%. For this study, 2,000 lb and 60 bbl per foot scenarios were not forecasted since it is not part of industry practice to use such a large completion job in the Williston basin.

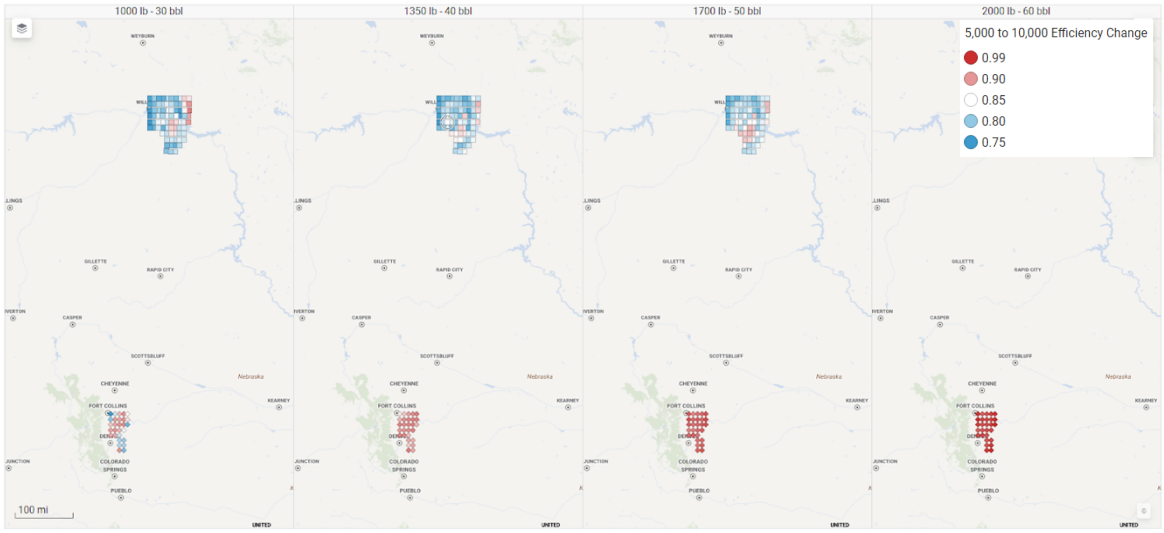


Figure 9. Lateral length efficiency for increasing from 5,000 ft to 10,000 ft in the DJ and Williston.

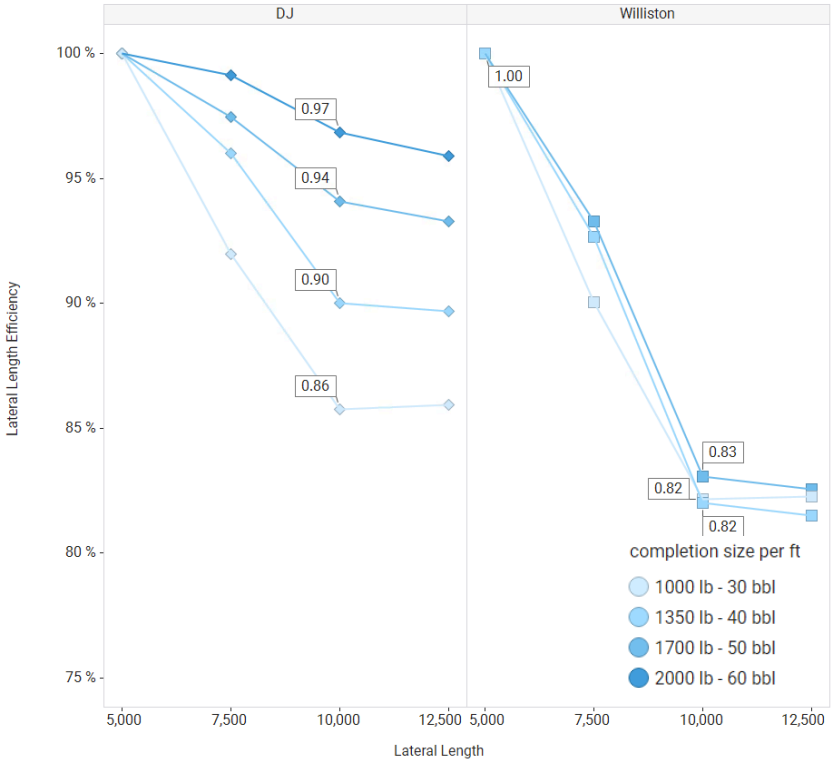


Figure 10. Lateral length efficiency curve for each completion design within the DJ and Williston.

In summary, the results from this paper do not show any basins with a 100% production efficiency across all LLs, but 2-mile-long wells in the Midland and EF-Gas Window have above a 95% efficiency. Interestingly Delaware and EF-Oil Window wells exhibit a similar steep decline in the lateral length efficiency curve. On the other hand, the wells in the Williston and DJ have steep drops, but the efficiency curve begins to exhibit an asymptotic behavior. Figure 11 summarizes the basin-wide change in production efficiency when increasing well lateral length from 5,000 ft to 10,000 ft.

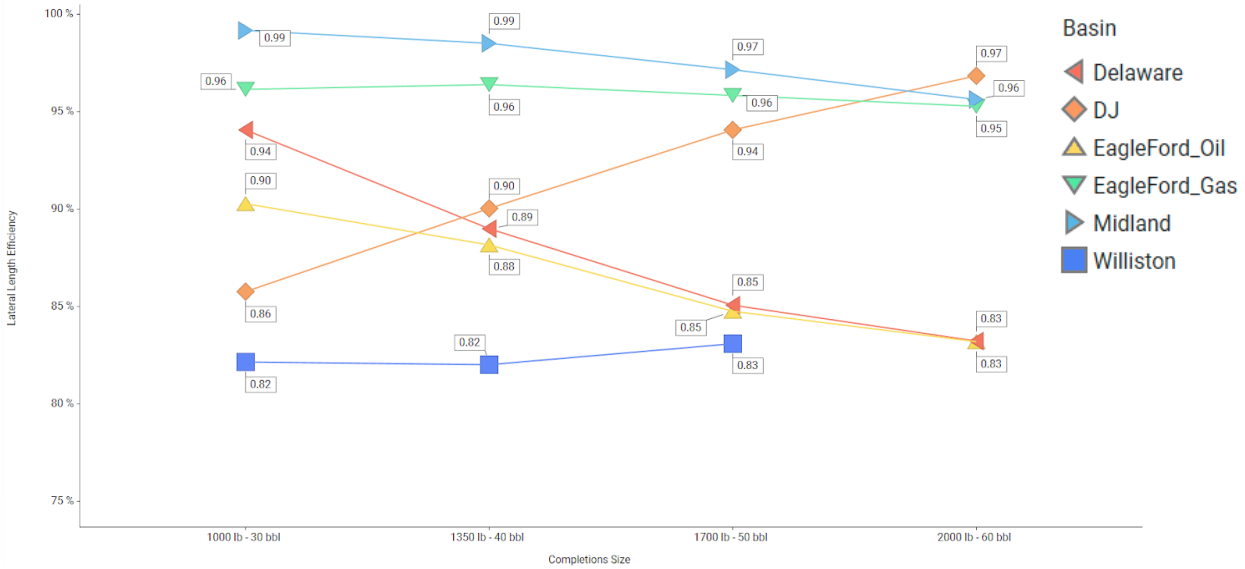


Figure 11. Lateral length efficiency vs completions size for increasing well lengths from 5,000 ft to 10,000 ft.

Discussion

Drivers of Lateral Length Inefficiency

Engineers consider various mechanisms by which longer laterals produce less efficiently than shorter laterals, and we note that these mechanisms are not mutually exclusive. We will discuss several prominent theories and their relation to the results of this study. First, the longer the lateral, the greater average distance of each producing foot from the artificial lift system, which may be thus on average less effective. Second, completions effectiveness may decline with increasing lateral length, as friction between the completions fluid and the casing accumulates. Third, longer laterals generally present a larger number of sumps or doglegs, through which production is more difficult. Finally, hydraulic fractures do not merely produce in two dimensions away from the wellbore; fractures develop in three dimensions and hydrocarbons do flow short distances to the fracture faces. In other words, the heel or toe of a lateral should be expected to produce hydrocarbons from some distance beyond the completed lateral; this "extra-lateral" drainage will diminish in relative importance to the total production with longer laterals.

We will review two key observations across the basins analyzed in this study, and their implications for the above hypotheses. First, the decrease in efficiency is more severe at shorter laterals. Generally, we see our models forecast a larger decline in per-foot production going from 5,000 to 7,500 feet, than going from 7,500 feet to 10,000 feet. Second, more intense completions generally have lower lateral length efficiency than smaller completions.

These observations are consistent with diminishing completions effectiveness with longer laterals, and with diminished lift effectiveness with longer laterals. If the completion at 10,000' in the lateral is significantly less effective than that at the heel, it stands to reason that intensified completions will offer

less uplift at 10,000' than at the heel. Lift effectiveness should have an asymptotic loss in efficiency with increasing lateral length – beyond a certain point, the artificial lift will have very limited impact at all, and thus increasing lateral length beyond that point will have very limited impact on the overall production efficiency.

Wellbore tortuosity, however, should have an increasing impact with length of the well, meaning we should observe a worsening production efficiency with length. This is because sumps and doglegs should increase in frequency and severity with length of the wellbore, as the trajectory becomes more and more difficult to control.

While this study did show a clear interaction between completions and lateral length efficiency, we did not directly examine interactions with artificial lift, wellbore geometry, and toe direction. Further analyses could incorporate these data types to investigate these relationships.

Ramifications for Future Developments

The "flattening" production efficiency degradation for longer laterals suggests that existing trends of longer lateral lengths may be expected to continue. We do note that additional data may change the model forecasts; the 12,500' and 15,000' lateral length space is less sampled than 5,000' to 10,000'. Furthermore, other technical considerations or operational risks in drilling and completions may make super-long laterals less attractive to operators.

While this study focused on the aggregate production efficiency of longer laterals, we note that the optimal length will vary based on geological conditions, unit spacing/stacking, and of course surface or unit development constraints; existing offset wells may dictate what interwell spacing and lateral length distances can be used for certain acreages. Thus, for each potential development unit, we recommend workers consider a range of lateral lengths simultaneously with different completions designs and development densities.

GOR Impacts

The gassier areas of most of the plays analyzed show greater lateral length efficiency. For instance, the core of the Wattenburg area shows greater lateral length efficiency than the other, less-gassy parts of the DJ. This is also true for the Eagle Ford Gas vs Oil Window and true to a lesser magnitude for the Williston Basin. This could be caused from the dissolved gas delivering more reservoir energy to wellbore and a lighter column of fluid in the vertical section. However, we did not observe any LL efficiency correlation to GOR in the Permian Basin.

Producing from Multiple Formations

Additionally, the analyses conducted in this paper only forecast the outcomes from developing a single formation at a time. In reality, oftentimes multiple formations are drilled and completed simultaneously. Creating additional hypothetical wells in other formations and following similar forecast steps will provide insights on differences when producing from multiple zones. However, since formation thickness and structural depth varies throughout the basin, inter-zone communication can be expected to vary from unit to unit.

Conclusions

In the vast majority of scenario comparisons, we do not observe a 100% lateral length efficiency; doubling lateral length from 5,000 ft to 10,000 ft does not produce double the oil production. In all cases

we found a continual decrease in lateral length efficiency as wells get longer. While some basins were observed to have a 95% lateral length efficiency, others could have a 20% reduction in 3 year per foot cumulative oil production. Furthermore, the magnitude of lateral length diminishing returns varies from one part of the region to another.

Lateral length efficiency is also dependent on completion size. In the Delaware, Eagle Ford-Oil Window, and Midland basins, the forecasts with larger completion sizes have worse lateral length efficiencies compared to the forecasts at the same locations with smaller completions sizes. While only the DJ basin shows the opposite trend; larger completions produce better lateral length efficiencies. The Eagle Ford and Williston basins seem to have minimal impacts from completion size.

Acknowledgements

The authors would like to thank Novi Labs for providing data and granting permission to publish this work. The authors also thank the data scientists and software engineers at Novi Labs for constructing the software platform.

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